

# PLS 152: Plant Genetics



Dr. J Grey Monroe ("Grey")



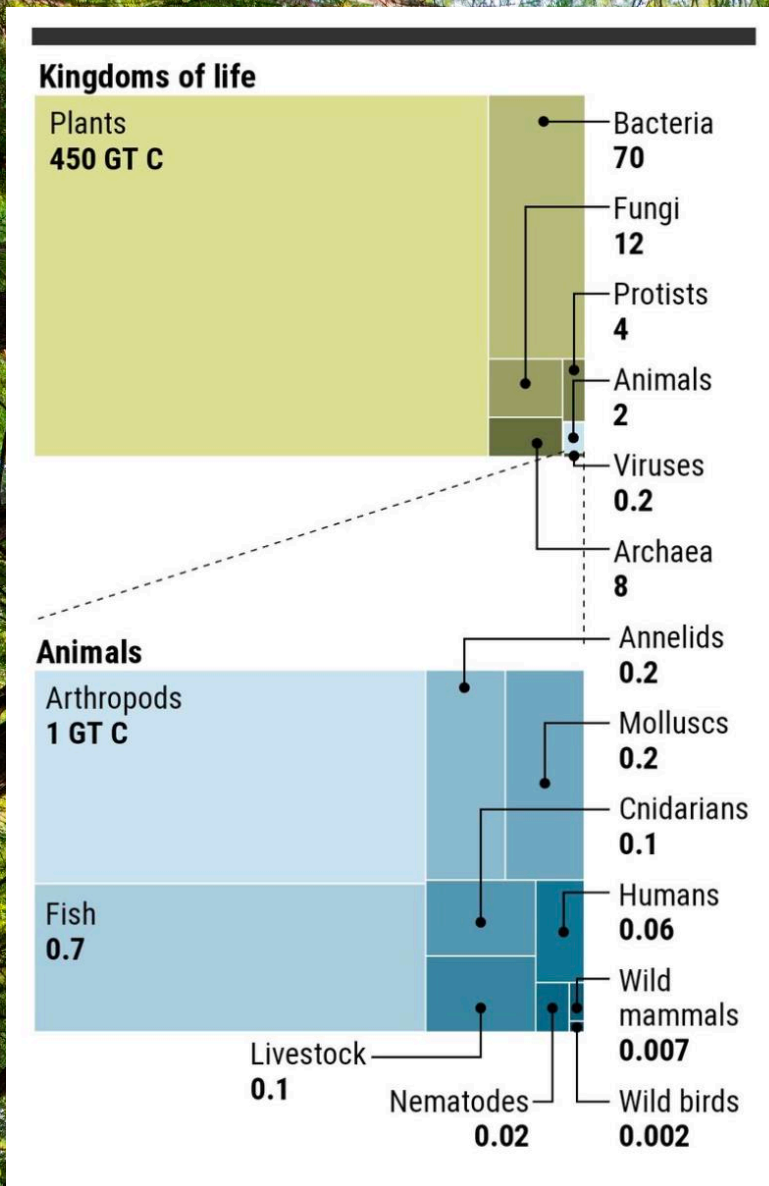


**Intro: preview of the course**

- A brief history of how plants are awesome and important
- Why understanding genetics is so valuable
- 9 fun facts about plant genetics
- Review of syllabus
- Goals for the course

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# 90 % of biomass on Earth is plants

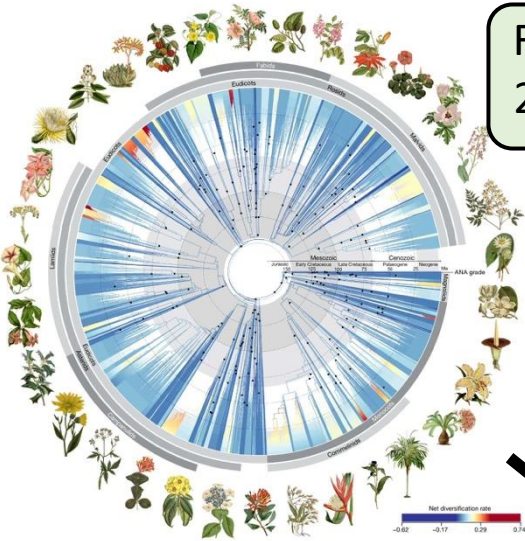
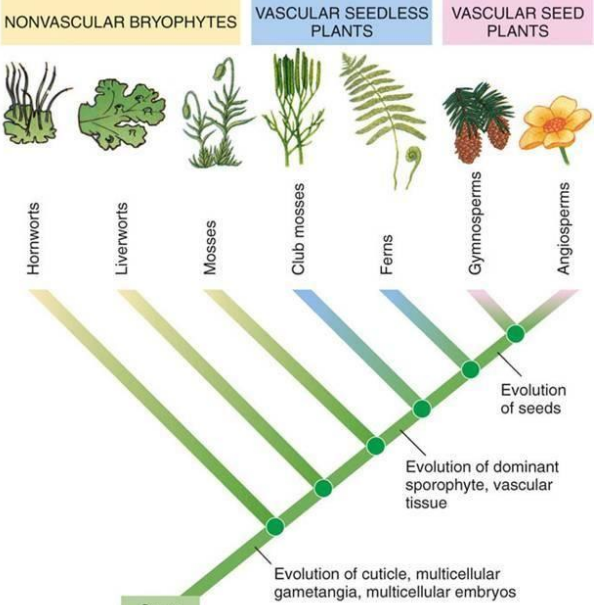


# Yearly cycle of photosynthesis measured from space



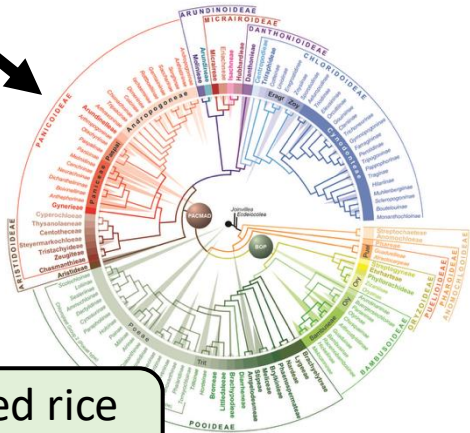
# How did plants evolve?

Land plants:  
500 Million years

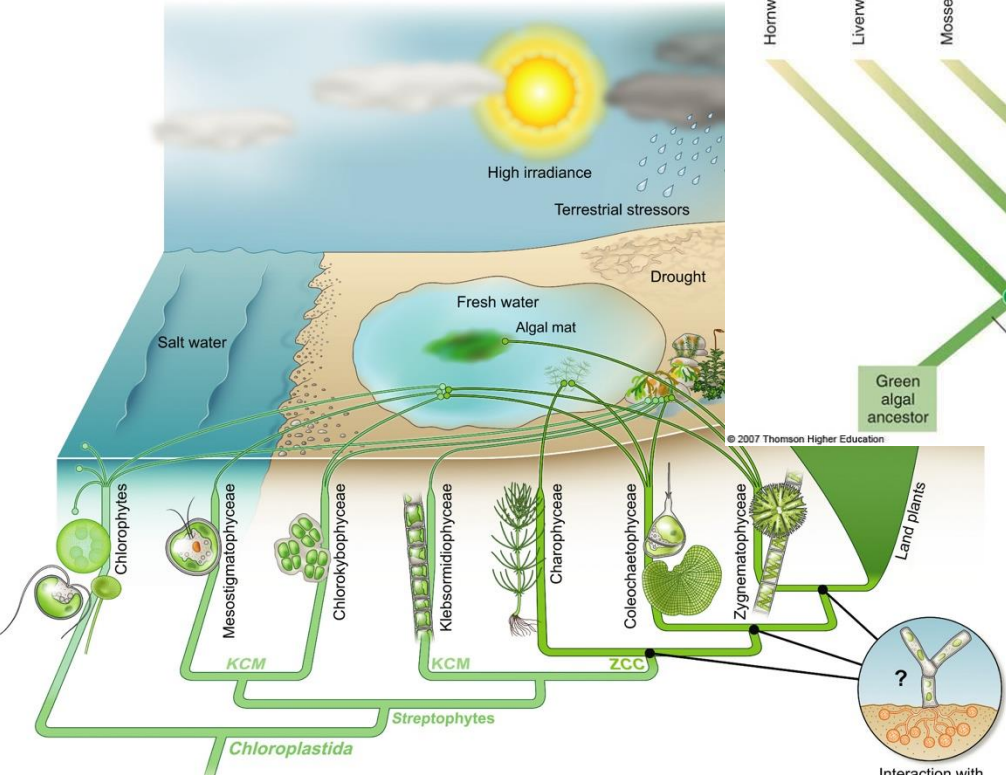
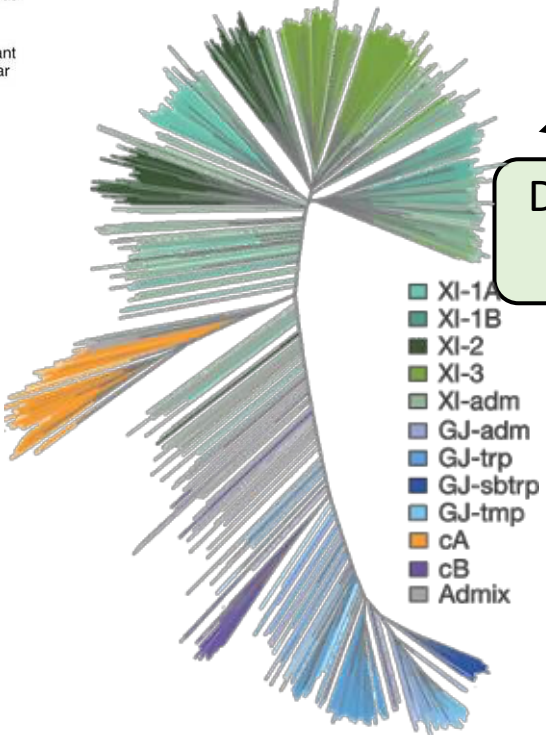


Flowering plants:  
200 Million years

Grasses:  
100 million years



Domesticated rice  
9,000 years ago



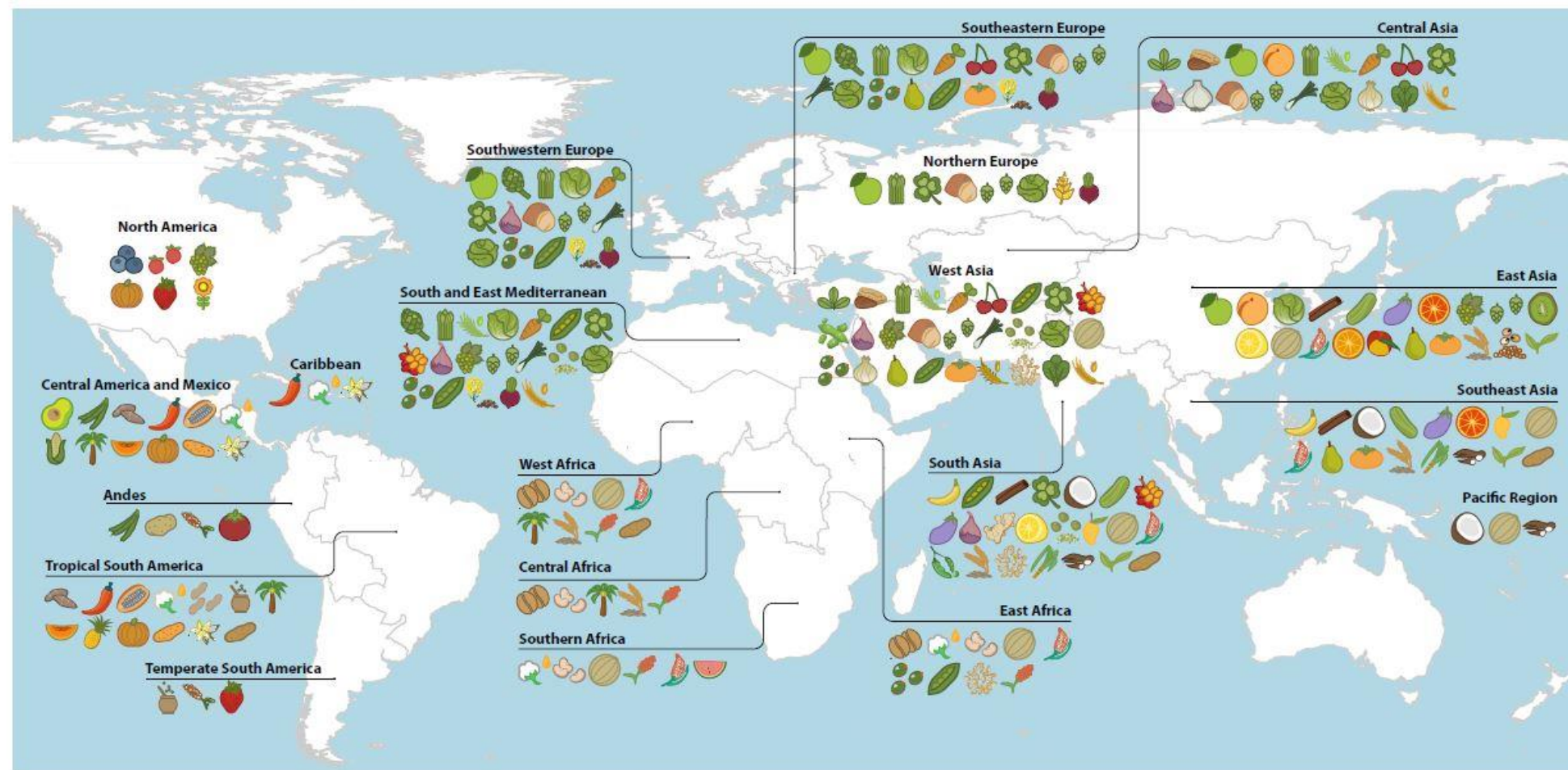
First "plant":  
1 Billion years ago...

# ORIGINS AND PRIMARY REGIONS OF DIVERSITY OF AGRICULTURAL CROPS



Khoury CK, Achicanoy HA, Björkman AD, Navarro-Racines C, Guarino L, Flores-Palacios X, Engels JMM, Wiersma JH, Dempewolf H, Sotelo S, Ramírez-Villegas J, Castañeda-Álvarez NP, Fowler C, Jarvis A, Rieseberg LH, and Struik PC (2016). Origins of food crops connect countries worldwide. *Proc. R. Soc. B* 283: 20160792. DOI: 10.1098/rspb.2016.0792.

International Center for Tropical Agriculture  
Since 1967 Science to cultivate change



- |                     |                    |                |            |                |                      |                     |              |                |
|---------------------|--------------------|----------------|------------|----------------|----------------------|---------------------|--------------|----------------|
| Alfalfa             | Beans              | Clover         | Eggplants  | Hops           | Melons               | Pears               | Rice         | Sunflower      |
| Almonds             | Blueberries        | Cocoa beans    | Faba beans | Kiwi           | Millet               | Peas                | Rye          | Sweet potatoes |
| Apples              | Cabbages           | Coconuts       | Figs       | Leeks          | Oats                 | Pigeonpeas          | Sesame       | Taro           |
| Apricots            | Carrots            | Coffee         | Garlic     | Lemons & limes | Olives               | Pineapples          | Sorghum      | Tea            |
| Artichokes          | Cassava            | Cottonseed oil | Ginger     | Lentils        | Onions               | Plums               | Soybean      | Tomatoes       |
| Asparagus           | Cherries           | Cowpeas        | Grapefruit | Lettuce        | Oranges              | Potatoes            | Spinach      | Vanilla        |
| Avocados            | Chickpeas          | Cranberries    | Grapes     | Maize          | Palm oil             | Pumpkins            | Strawberries | Watermelons    |
| Bananas & plantains | Chillies & peppers | Cucumbers      | Groundnut  | Mangoes        | Papayas              | Quinoa              | Sugar beet   | Wheat          |
| Barley              | Cinnamon           | Dates          | Hazelnuts  | Mate           | Peaches & nectarines | Rape & mustard seed | Sugarcane    | Yams           |

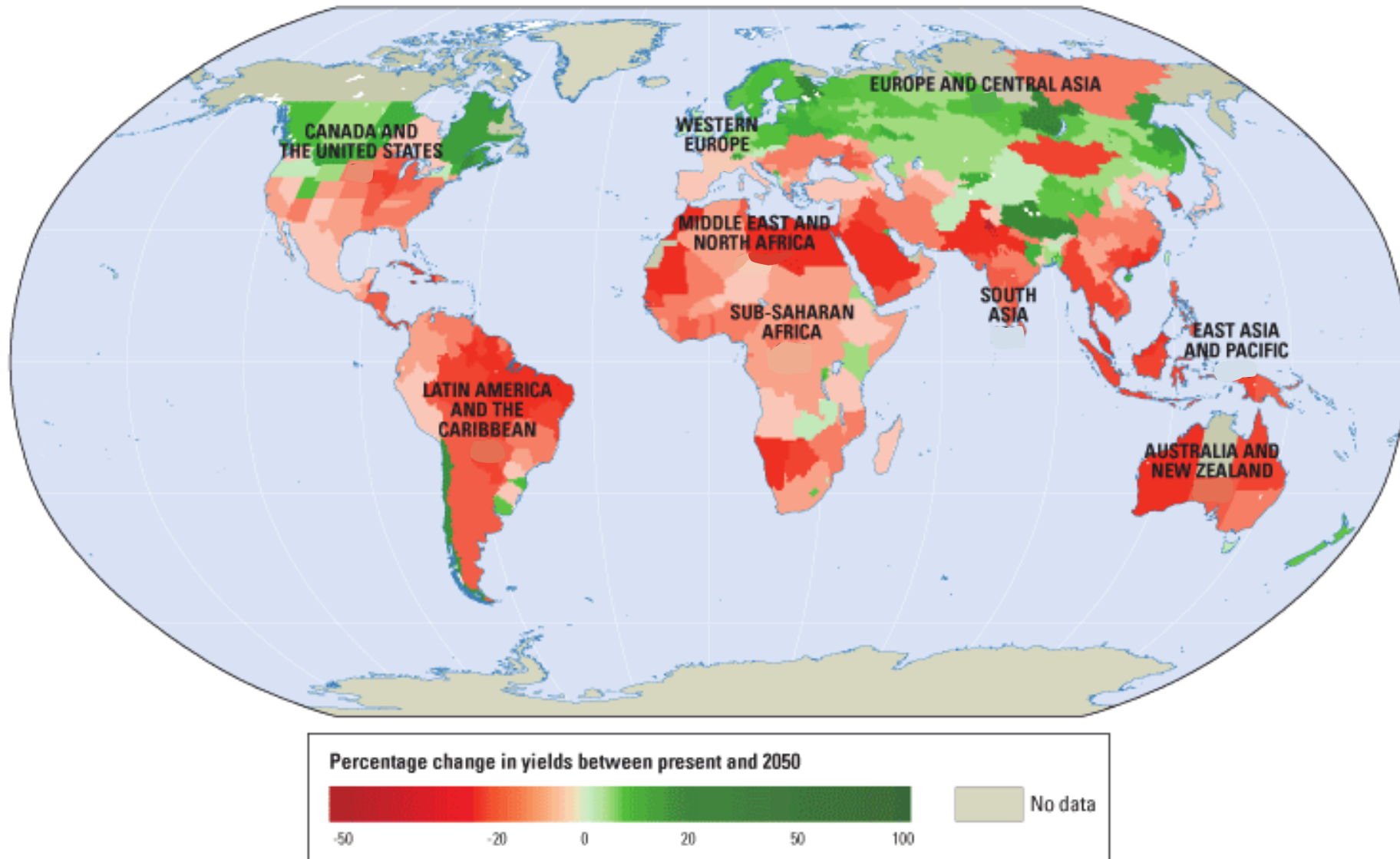
# Plants provide 90% of human calories

- Nutrients: Water, Carbohydrates, Fiber, Protein, Fat, Vitamins, Minerals, Secondary metabolites.
- There are 391,000 documented plants, 30,000 are edible, 5538 are counted as human food plants globally, but only 3 provide 50% of plant calories.



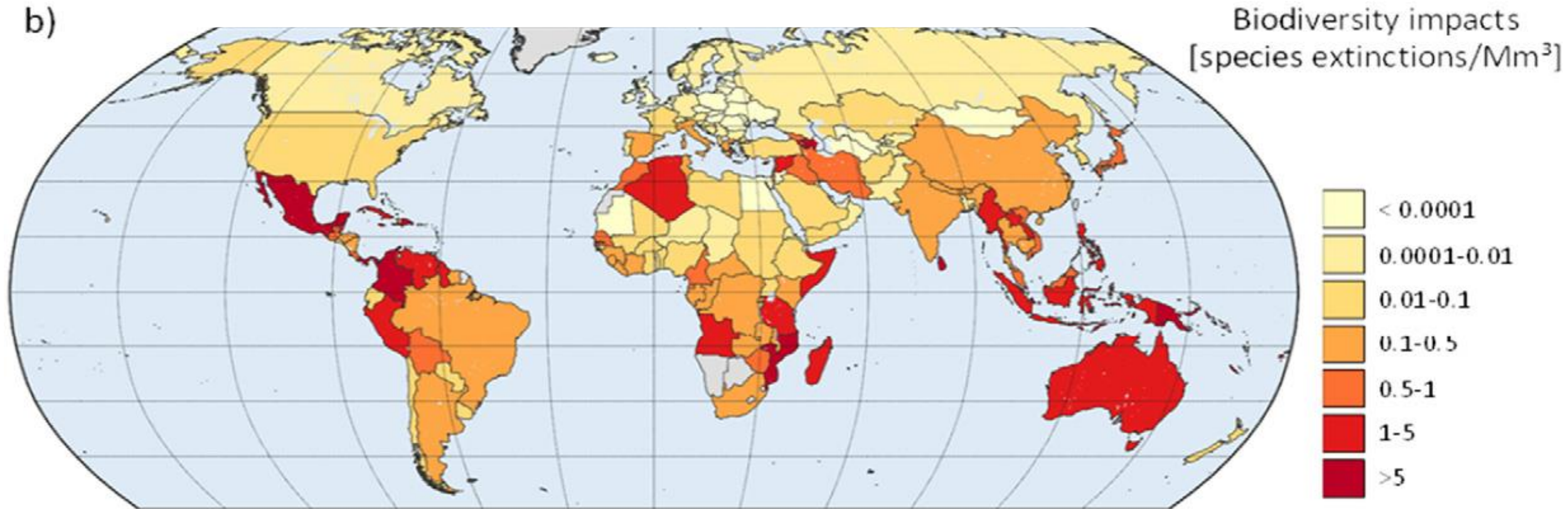
|    |              |       |
|----|--------------|-------|
| 1  | Maize Corn   | 19.5% |
| 2  | Rice         | 16.5% |
| 3  | Wheat        | 15.0% |
| 4  | Cassava      | 2.6%  |
| 5  | Soybeans     | 2.1%  |
| 6  | Potatoes     | 1.7%  |
| 7  | Sorghum      | 1.2%  |
| 8  | Sweet Potato | 0.6%  |
| 9  | Yams         | 0.4%  |
| 10 | Plantain     | 0.3%  |

# Climate change threatens crop yields...



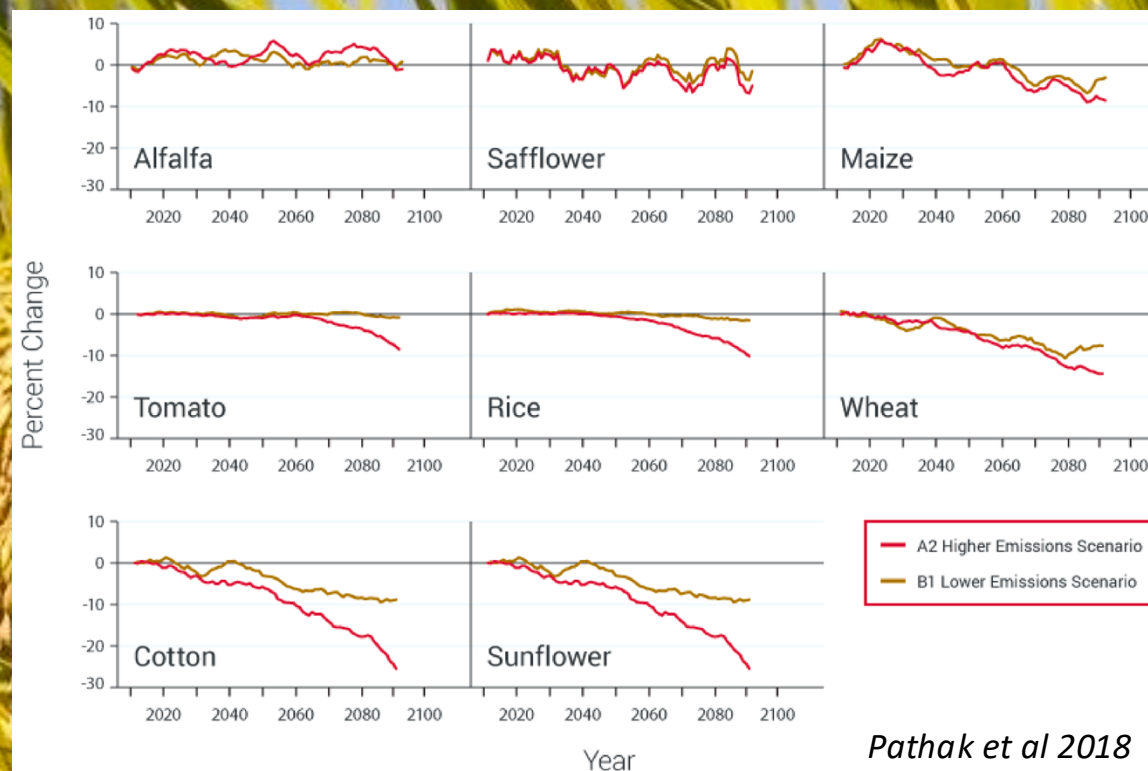
# ... and biodiversity

b)

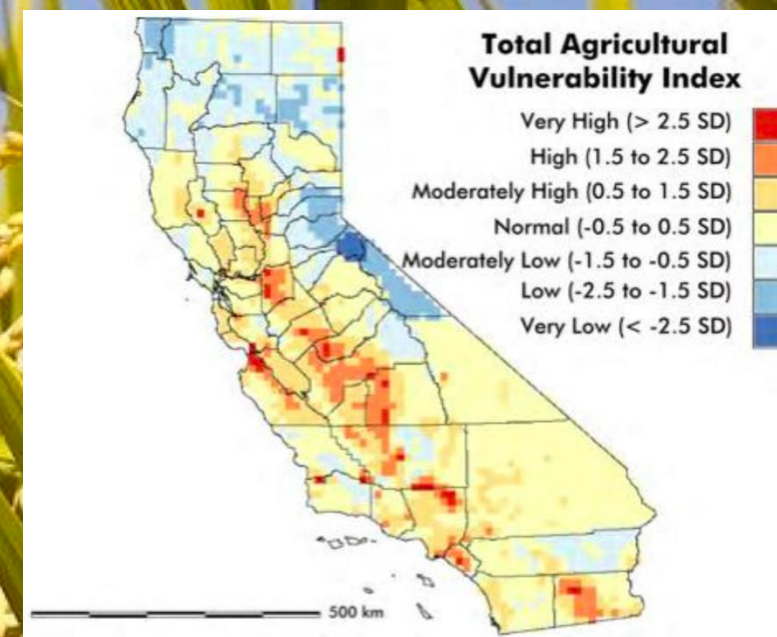


Tulare, Fresno and Kern Counties are among California farmers most affected by crop losses due to heat, totaling \$3 billion over the last 20 years

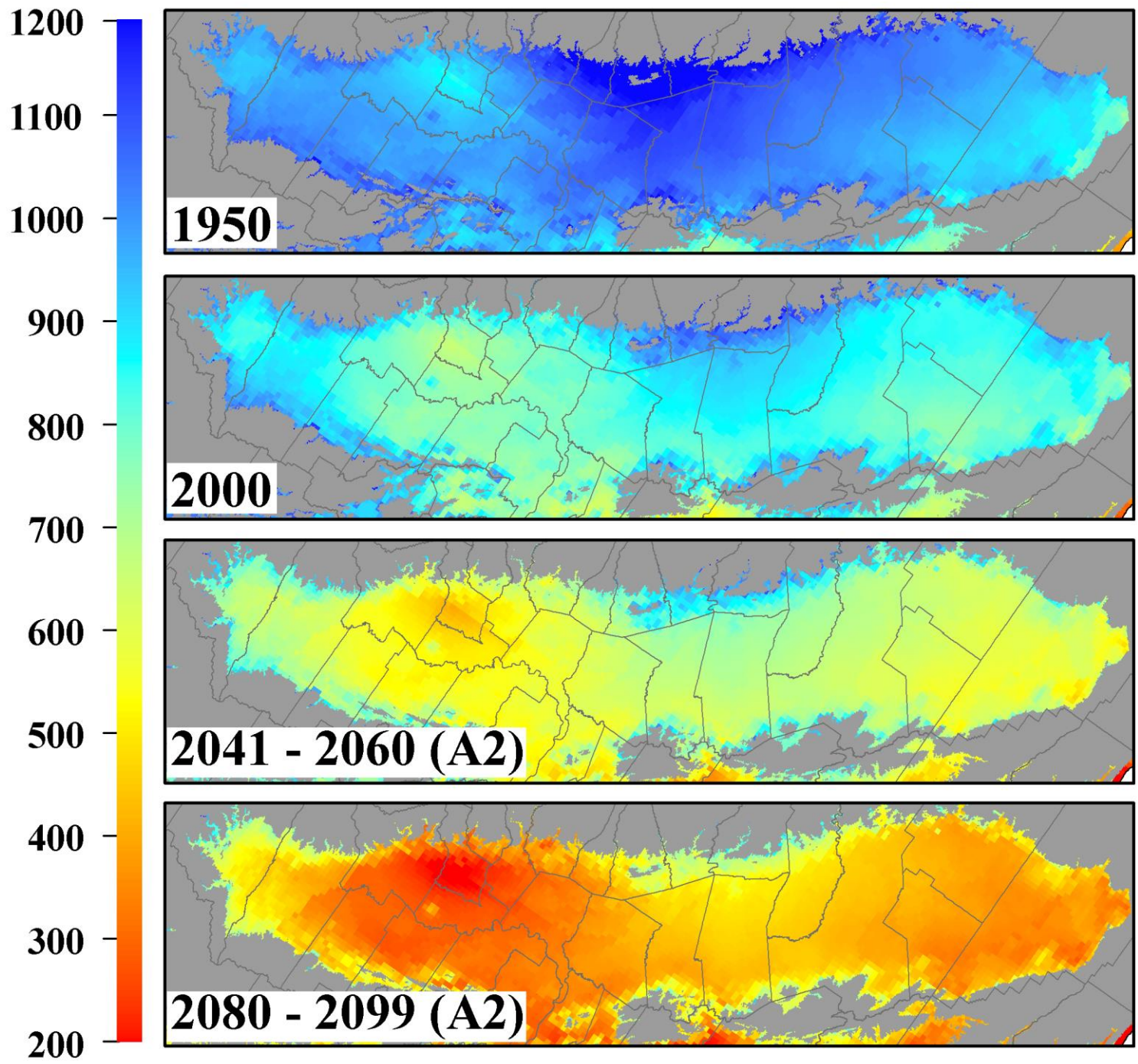
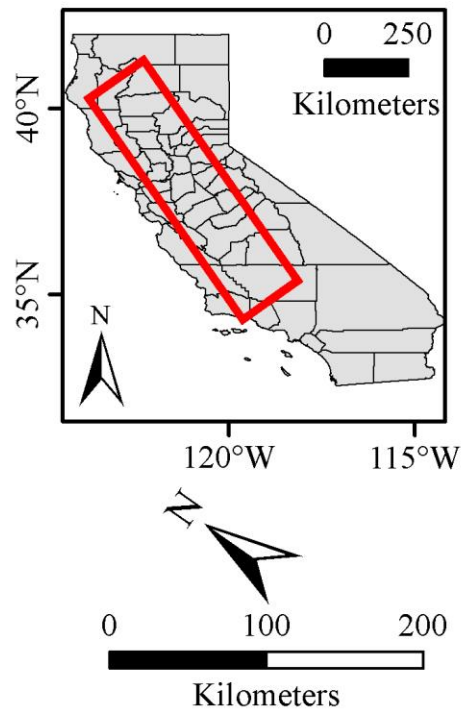
Lindt, 2019



Pathak et al 2018



# Safe winter chill (Chilling Hours)

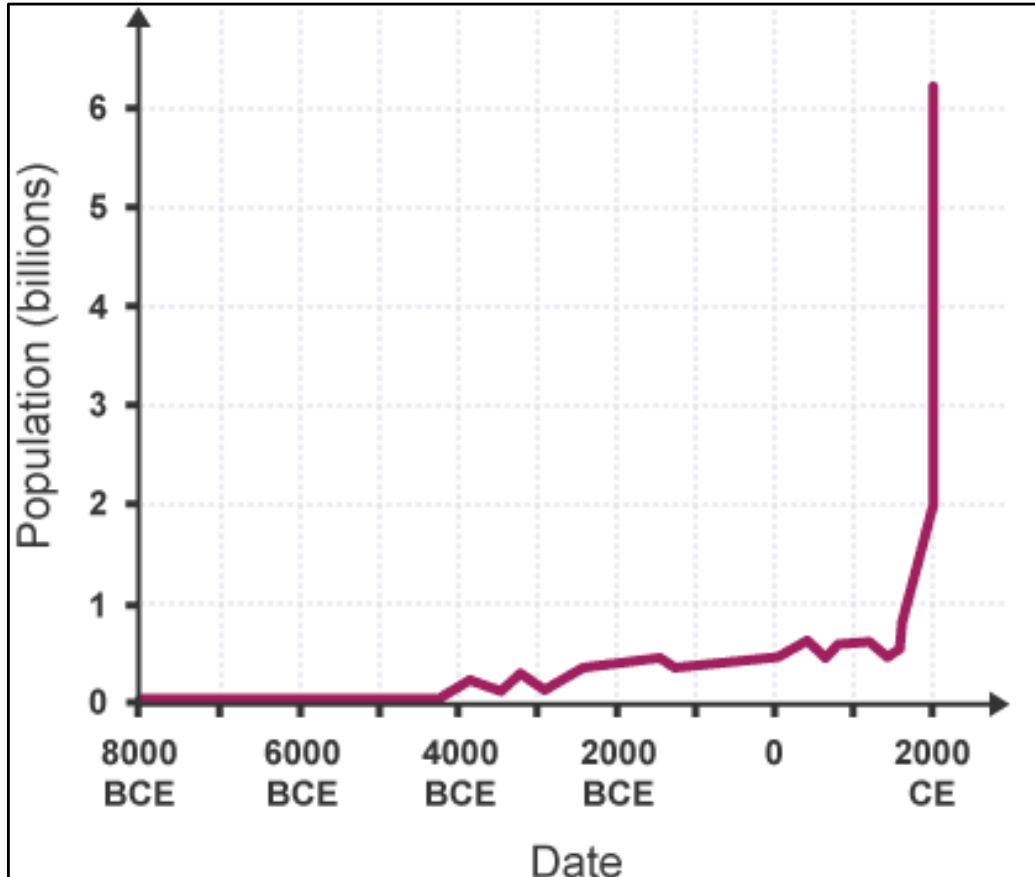


Sad pistachio

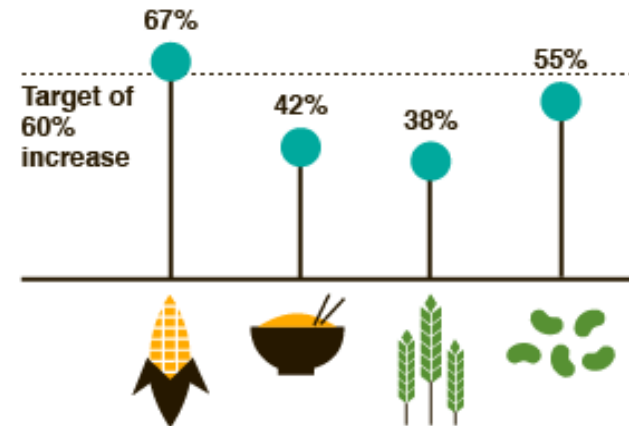


Luedeling et al. 2009

# Can current plant productivity meet our growing needs/problems?



Yields of maize, rice, wheat, and soybean all need to **INCREASE BY 60%**, by 2050 to meet demand but current growth in yield are falling short of the target.



**Big Facts**

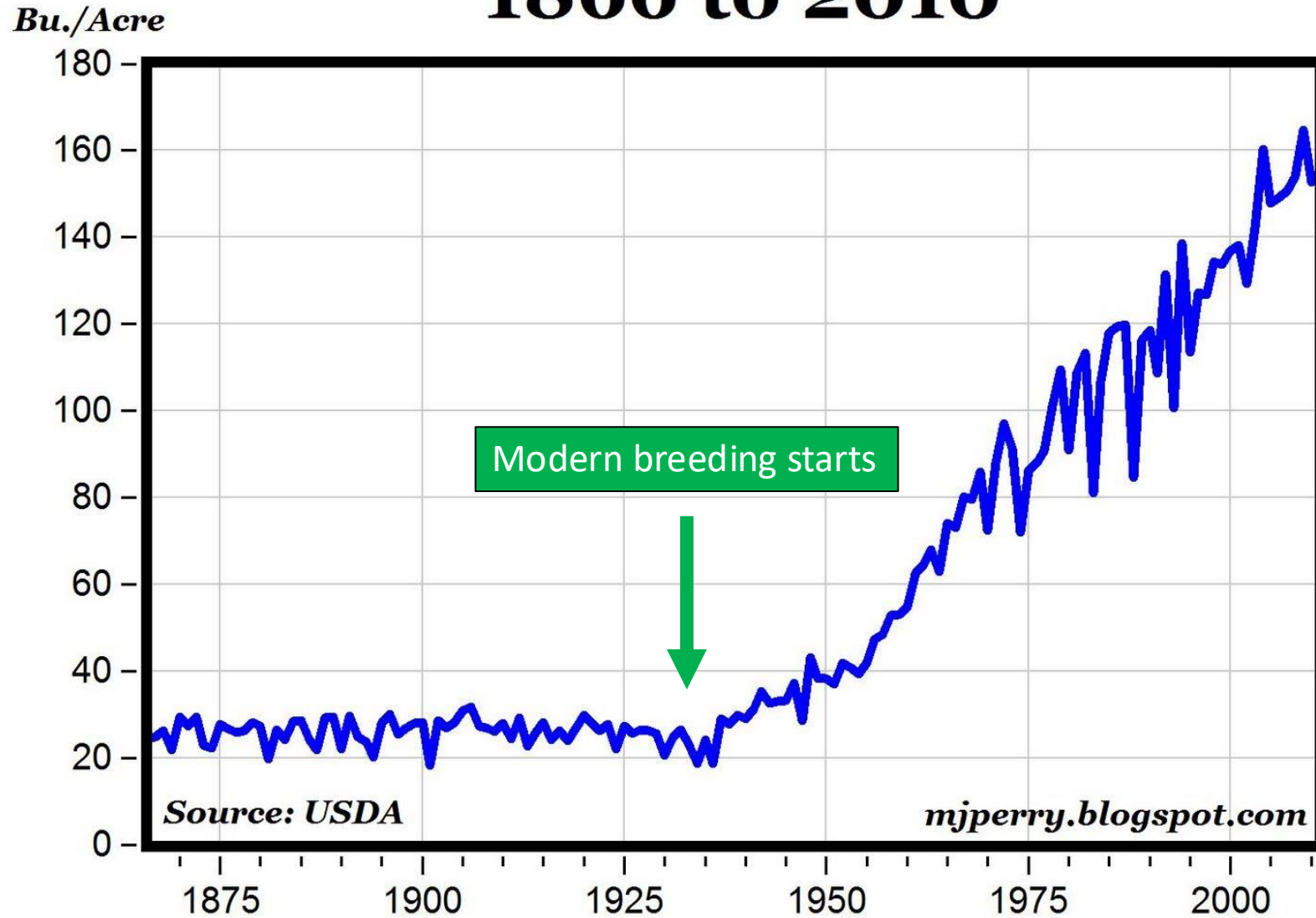
[ccafs.cgiar.org/bigfacts](http://ccafs.cgiar.org/bigfacts)



RESEARCH PROGRAM ON  
Climate Change,  
Agriculture and  
Food Security



# U.S. Corn: Yield per Acre 1866 to 2010





1. The potential of microbiomes--primarily in the animal gut and in soil--to increase efficiency and overcome obstacles in production
2. Advancements in genetic evaluation and editing, including making the most of CRISPR and other technologies to accelerate the evolution of food production
3. Expanding and analyzing many pools of data involved in growing and producing food
4. Developing and improving sensors and biosensors across all agricultural sectors to increase productivity and better target interventions
5. Examining, through transdisciplinary collaborations, entire systems in food production and finding the keys to adapting and transforming them to overcome challenges and increase production

Published July 13<sup>th</sup> 2018

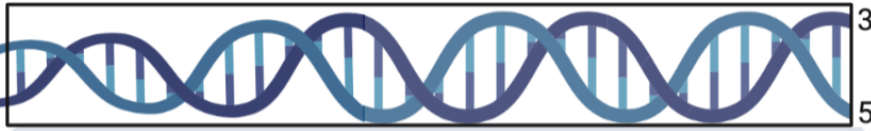
**Improved crops, resilient to  
environmental stress, requiring  
low inputs and with optimal  
nutritive value are needed**

**Requires a fundamental  
understanding of crop/plant  
genetics**

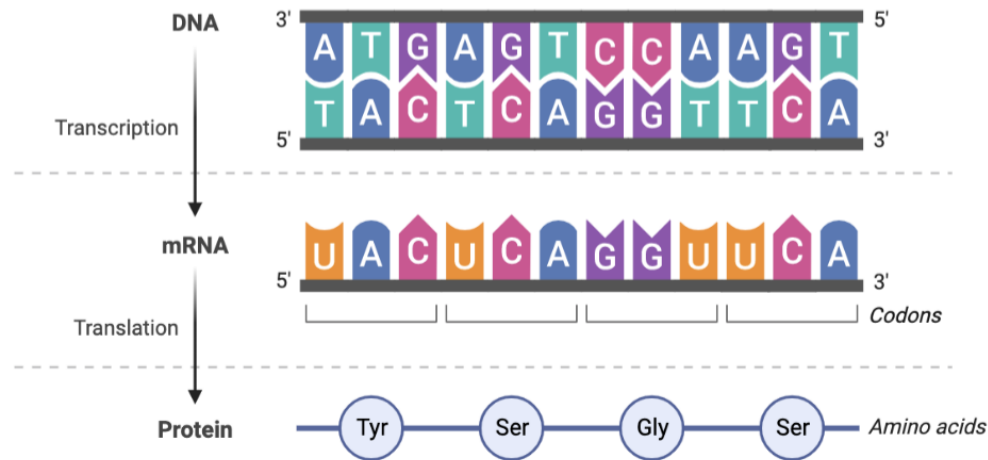
Chromosome



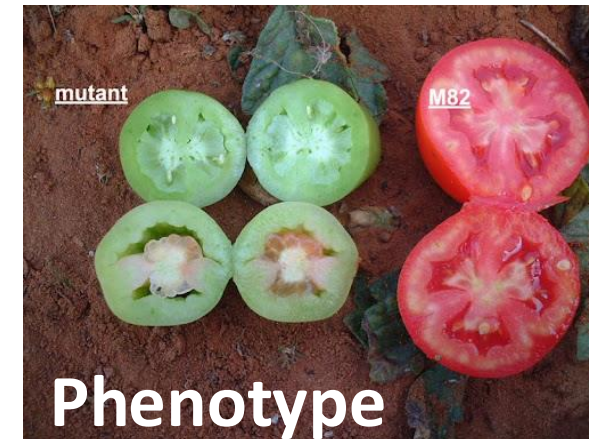
The genome is inherited by offspring



encodes the phenotype

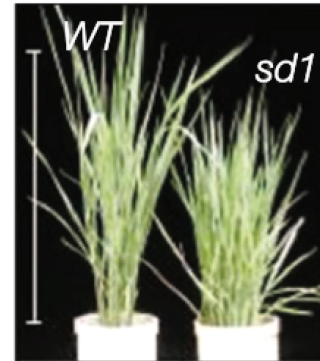
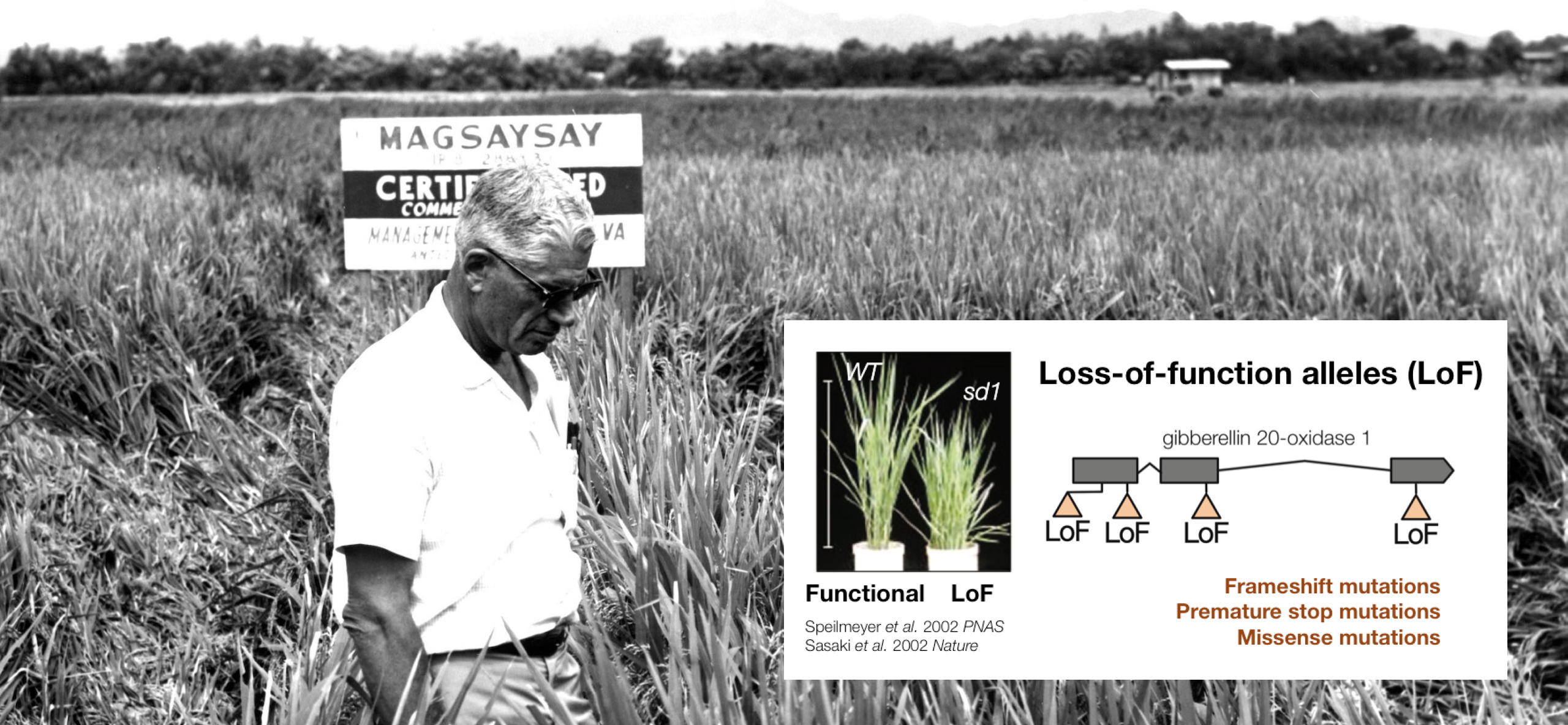


differences in the genome can produce different phenotypes



10 facts about plant genetics

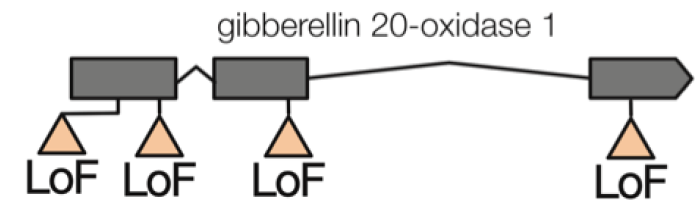
# A single mutation may have saved 100's of millions of people from starvation



**Functional LoF**

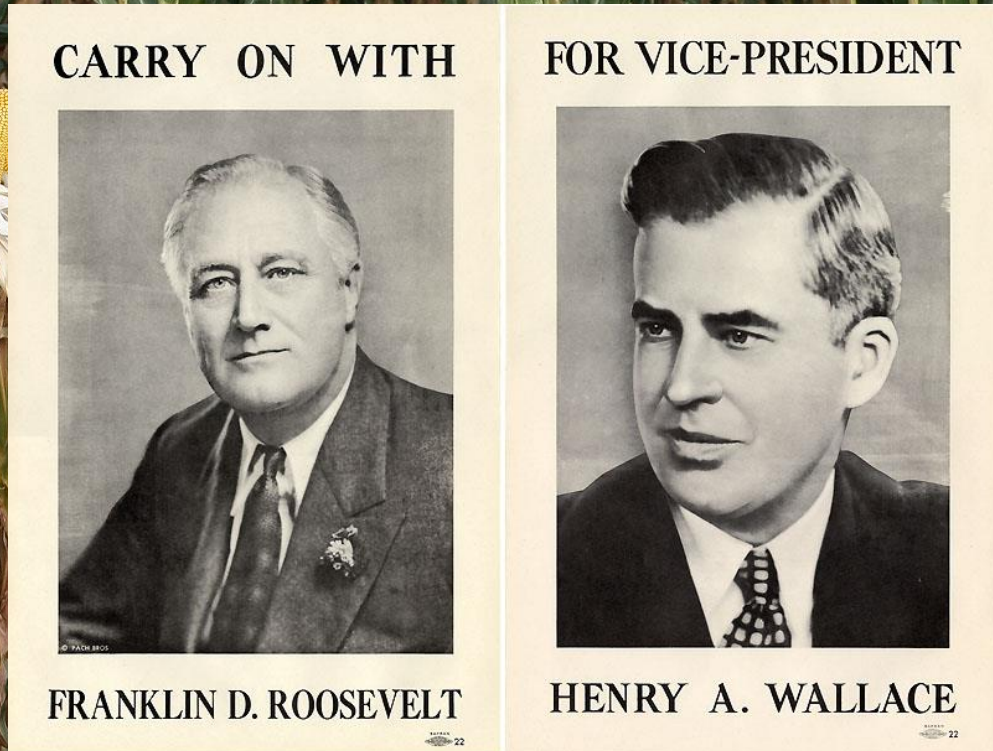
Speilmeyer *et al.* 2002 *PNAS*  
Sasaki *et al.* 2002 *Nature*

## Loss-of-function alleles (LoF)



**Frameshift mutations**  
**Premature stop mutations**  
**Missense mutations**

**F.D.R.'s vice president Henry Wallace of Progressive Party  
started *Pioneer* to produce hybrid corn ...  
still major breeding company today**

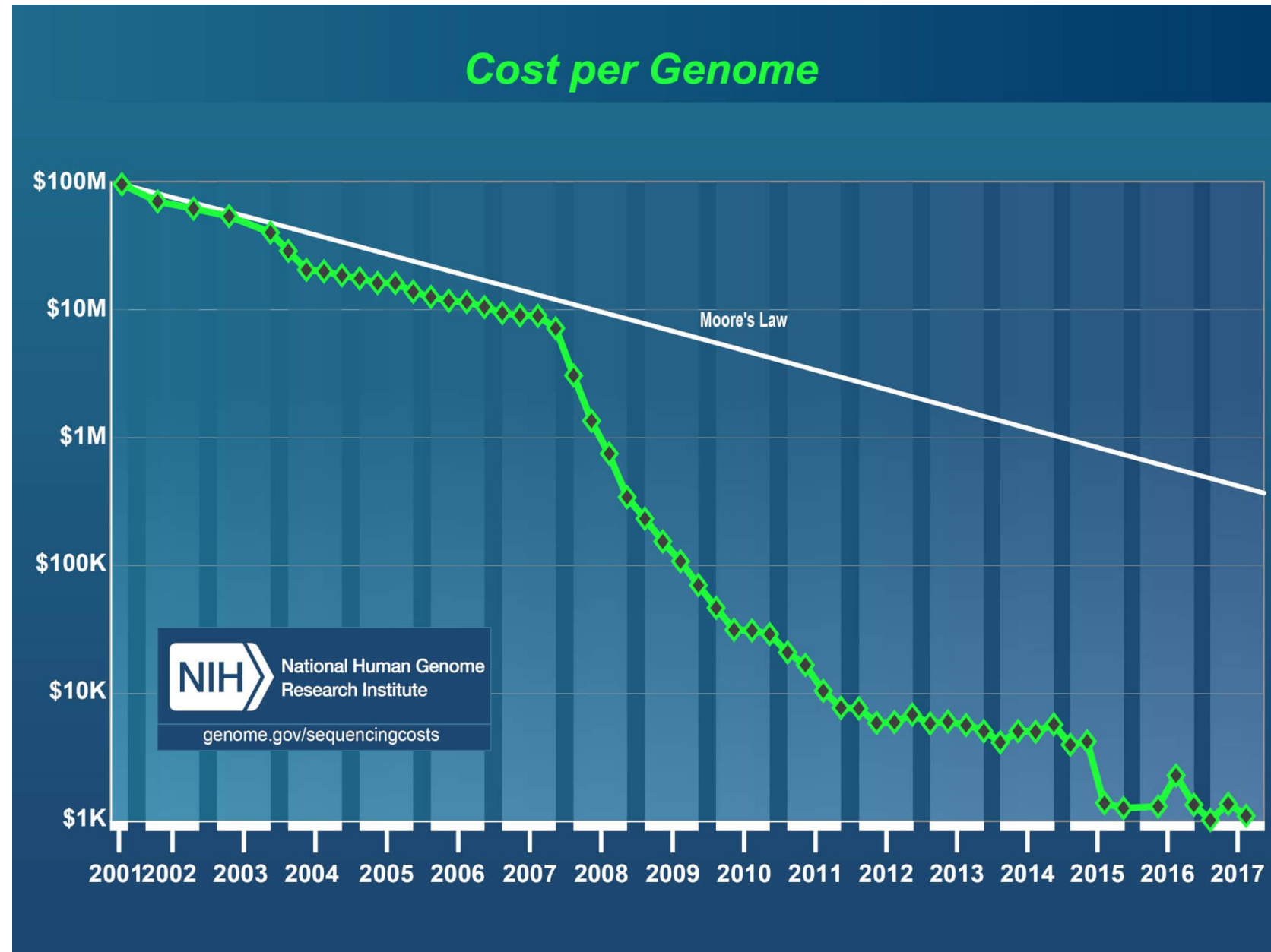


First plant genome  
sequenced cost  
**US\$100 million**

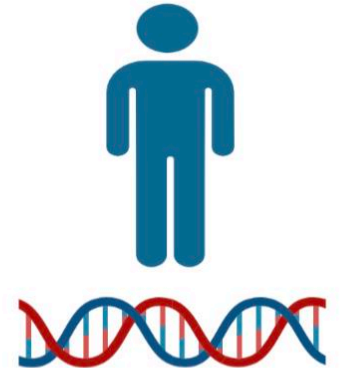
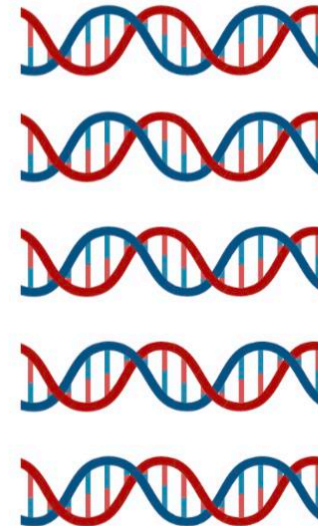
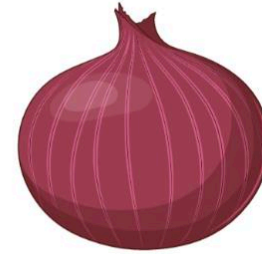
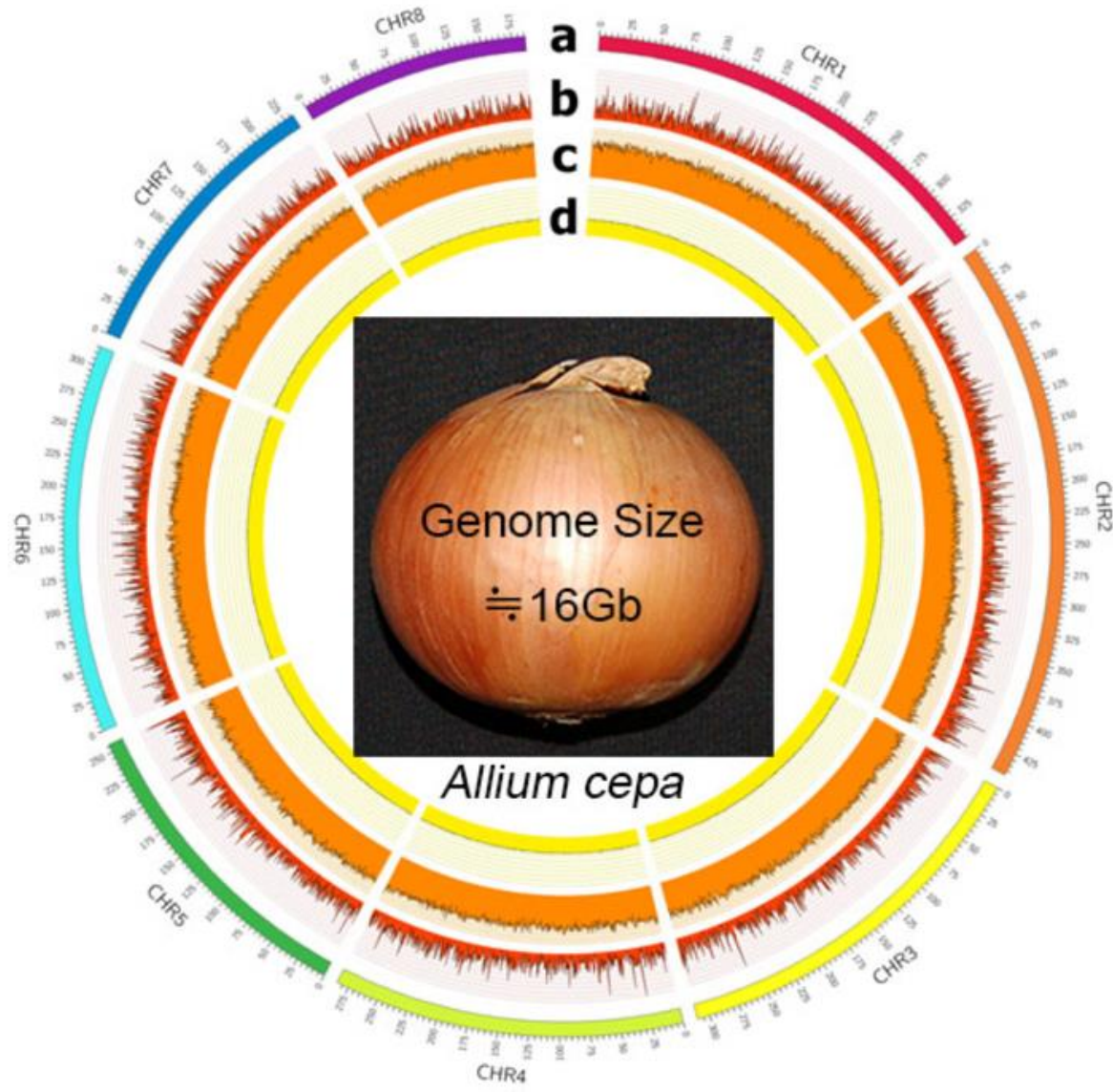
*Arabidopsis thaliana*



Today a genome of the  
same plant would be as  
little as **US\$100.00**



# Onions have 5 times more DNA than humans



# Some ferns have 40 times more chromosomes than humans



*Ophioglossum reticulatum*

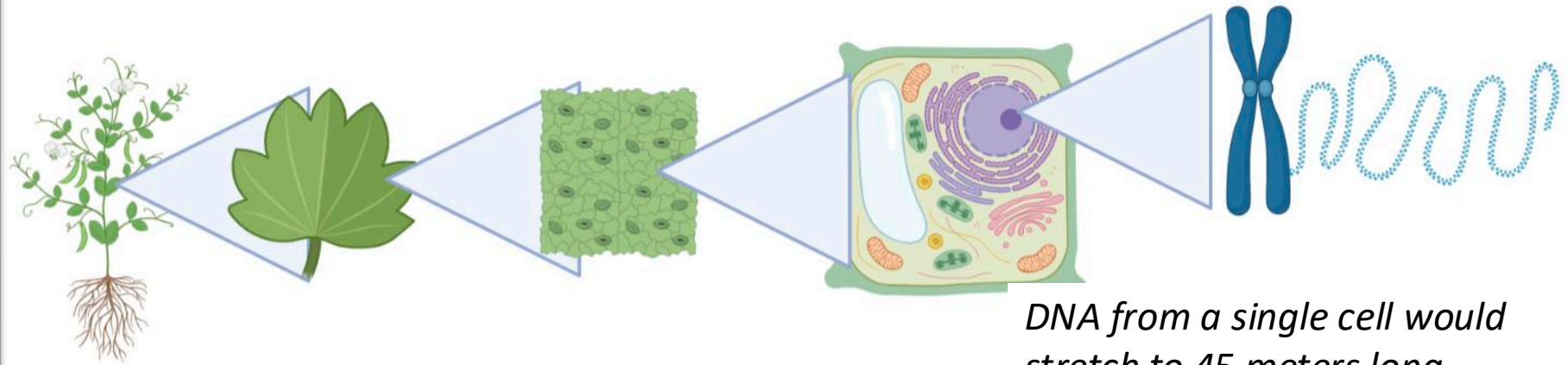


x46

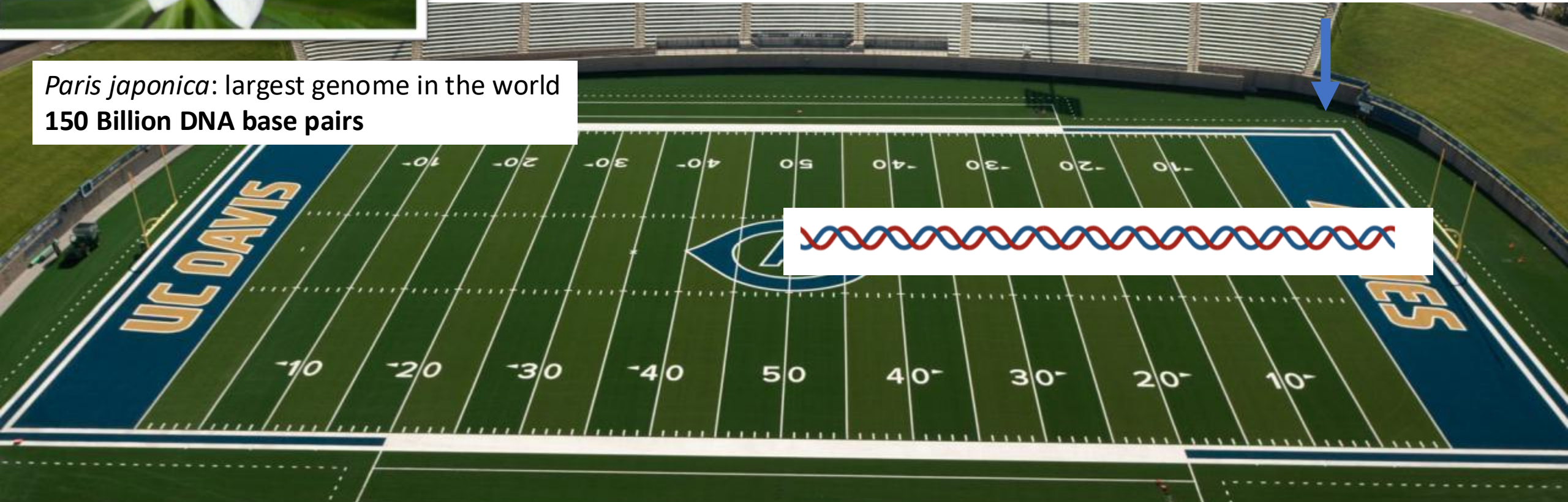


x1200

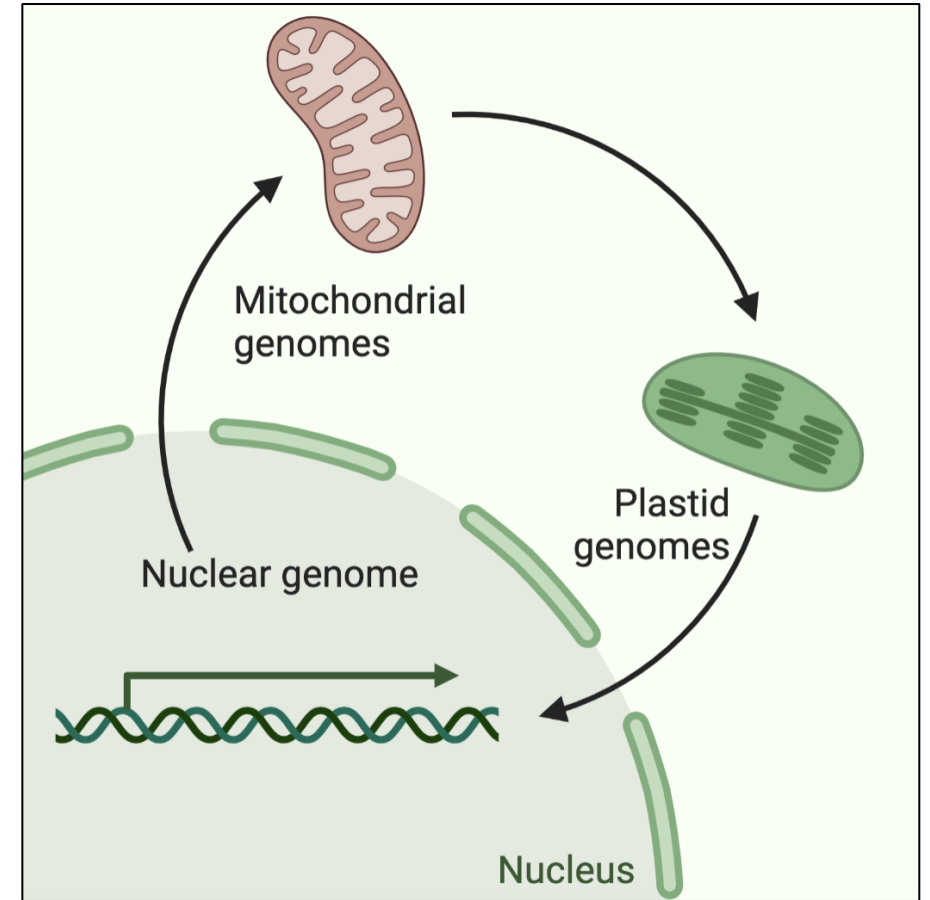
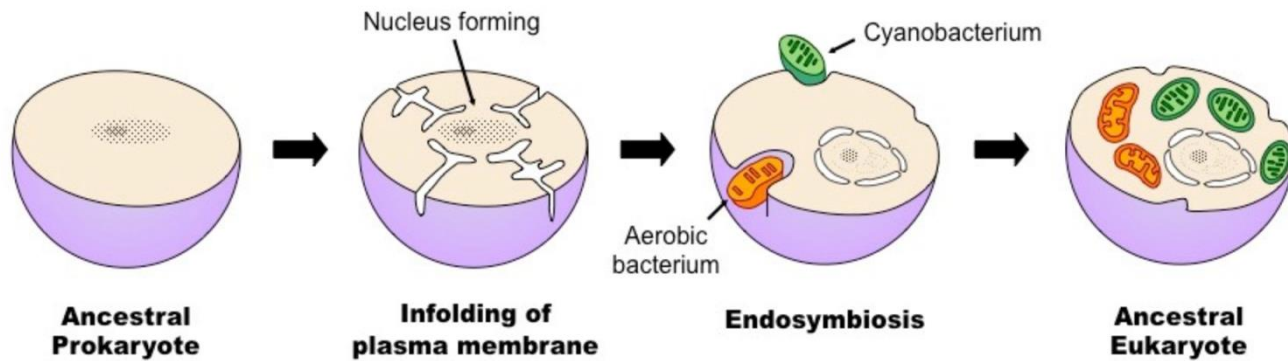
# The multicellular organism with the largest known genome is a plant



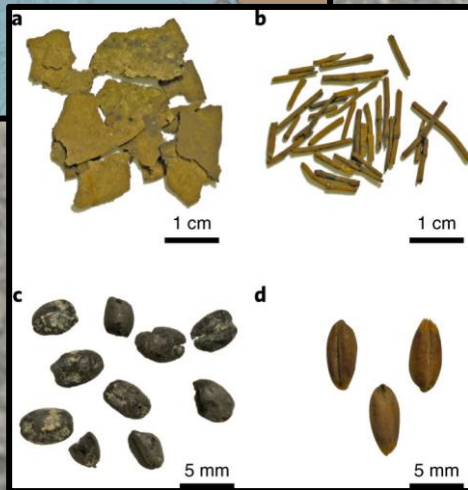
*Paris japonica*: largest genome in the world  
**150 Billion DNA base pairs**



# Plant cells have genomes originating from three organisms



# Ancient DNA has been collected from plants >1 million years old



# Many plants can genetically clone themselves



30 million pistachio clones in California

First “genetically engineered” crop was a tomato in 1994: *Flavr Savr*



Humans have been “genetically modifying” plants for thousands of years through breeding

## **A visual guide to identifying genetically modified corn**



**NON GM**



**GM**

Wild ancestor of corn

Modern corn

The central rules of genetics were discovered  
in plants



## Course materials:

### Syllabus

**PLS 152: Plant Genetics, Fall 2026** · 4 units · CRN 42467

Tuesdays and Thursdays 3:10–5:00 PM, Veihmeyer Hall 212

**Dr. Grey Monroe** — office hours Thursdays 10:00–11:00 AM, Robbins Hall 262A, or by appointment ([zoom](#)) · [gmonroe@ucdavis.edu](mailto:gmonroe@ucdavis.edu) (please allow 24 hours for reply)

**TA: Alice Pierce** — [avpierce@ucdavis.edu](mailto:avpierce@ucdavis.edu) · office hours by appointment (announced week 1). Alice is a PhD candidate in the Monroe Lab and Plant Biology Graduate Program, and a familiar face in this course: she was the PLS 152 TA in 2022.

### Course information

The primary aim of this course is to help you develop a solid foundation in basic genetics so that you will be able to apply this knowledge to understand how and

On this page

[Course information](#)

Course format: in person

Learning outcomes

Prerequisites and materials

Assignments and assessment

Grading policy

Use of AI

Schedule

Academic integrity

Accessibility

Student wellness

|              |                    |                       |            |   |                               |
|--------------|--------------------|-----------------------|------------|---|-------------------------------|
| <b>42467</b> | <b>PLS 152 001</b> | <b>Plant Genetics</b> |            | 4 | <b>J. Monroe</b>              |
|              | 3:10 PM - 4:30 PM  | T,R                   | Lecture    |   | Veihmeyer Hall 212            |
|              | 4:40 PM - 5:00 PM  | T,R                   | Discussion |   | Laboratory Veihmeyer Hall 212 |

One class with a break in between

## **ASSIGNMENTS & ASSESSMENT**

**Class participation (10%):** a short reflection on each class, submitted on Canvas and graded for completion, due by 11:59 PM the same day. Two questions, every time: what interested you most today, and what would you benefit from a deeper explanation of? I read these, and they shape the following lectures.

**Weekly assignments (30%):** low-stakes quizzes on Canvas covering the previous week's material, due Tuesdays at 3:10 PM, before class. Each quiz includes at least one question drawn from that week's research article. The lowest weekly assignment score is dropped automatically.

**Midterm (30%):** a written mini literature review on a plant genetics topic of your choice — maximum 1,500 words, at least 5 peer-reviewed sources, due Tuesday November 10 at 3:10 PM. The October 15 library session launches your literature search, and the Topic Explorer (due October 22) helps you pick a topic with evidence instead of guesswork. Graded anonymously against a rubric posted with the assignment.

**Capstone project (30%):** the final for this class is a group presentation (5–10 minutes) exploring a plant genetics topic — presentations run December 1 and 3. Groups form from a preference survey (due October 29), with two short check-ins along the way. Graded on a rubric covering content and quality, plus each member's individual contribution statement.

**Extra credit (up to 2%):** memes! Post an original genetics meme to the class gallery by Friday December 11 — 1 point, or 2 if it's plant genetics specific. A proud course tradition.

## GRADING POLICY

Your success in this class is important to me. We will all need accommodations because we all learn differently. If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. Together we'll develop strategies to meet both your needs and the requirements of the course.

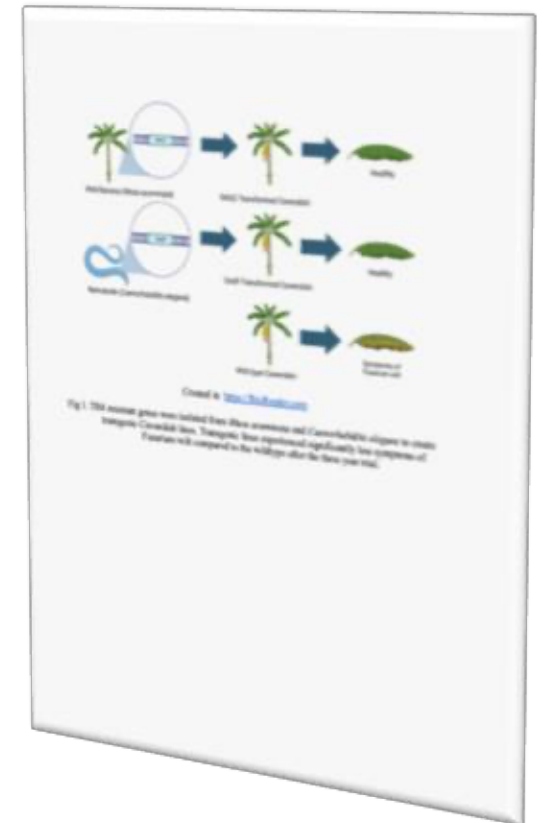
**“Time Banks” late assignment policy:** students have a two-day grace period for one assignment, or two one-day extensions for two different assignments. Additional late work is eligible for 60% of original points. Time Banks do not apply to the final project (a scheduled presentation) or to daily reflections — a reflection on today's class only makes sense today; completion grading and drop-lowest are the built-in slack there. For extraordinary circumstances (e.g. illness, emergency), please contact me as soon as possible to make arrangements.

Talk with me as early in the quarter as possible about any accommodations you need. Contact the Student Disability Center for more information and to request accommodations: (530) 752-3184 · [sdcc@ucdavis.edu](mailto:sdcc@ucdavis.edu) · [sdcc.ucdavis.edu](http://sdcc.ucdavis.edu). There are also a range of resources on campus, including the Writing Center, Tutoring Center, and Academic Advising Center.

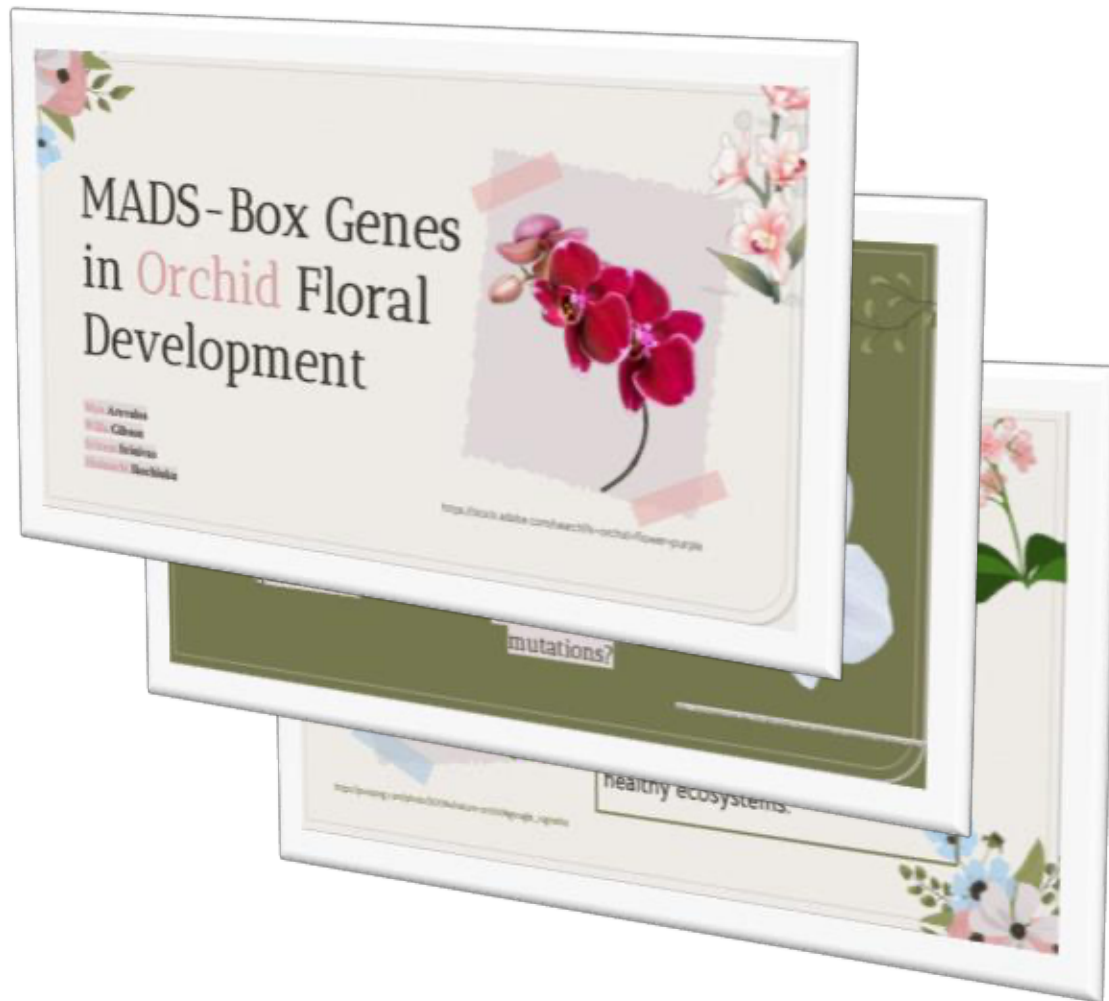
There will be opportunities for extra credit, and students are encouraged to take advantage of these as they are available.

| Week                 | Class     | Date                         | Time            | Lecturer                                  | Topic                                                                                                                 | Due before class:                                                      |
|----------------------|-----------|------------------------------|-----------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 0                    | 1         | Thursday, September 24th     | 3:10pm - 5:00pm | Grey                                      | Intro: Why plant genetics is awesome                                                                                  | ----                                                                   |
| 1                    | 2         | Tuesday, September 29th      | 3:10pm - 5:00pm | Grey                                      | Transmission Genetics 1: Mendel's Rules                                                                               | Reflection                                                             |
| 1                    | 3         | Thursday, October 1st        | 3:10pm - 5:00pm | Grey                                      | Transmission Genetics 2: Breaking Mendel's Rules                                                                      | Reflection                                                             |
| 2                    | 4         | Tuesday, October 6th         | 3:10pm - 5:00pm | Grey                                      | Molecular genetics: Mutations – the ultimate source of all genetic variation                                          | Reflection & <b>Quiz 1</b> (Transmission genetics)                     |
| 2                    | 5         | Thursday, October 8th        | 3:10pm - 5:00pm | Grey                                      | Molecular genetics: how does an allele cause a phenotype                                                              | Reflection                                                             |
| 3                    | 6         | Tuesday, October 13th        | 3:10pm - 5:00pm | Grey                                      | Importance of Reproduction for plant genetics / Domestication: the origins of crops                                   | Reflection & <b>Quiz 2</b> (Molecular genetics)                        |
| 3                    | 7         | Thursday, October 15th       | 3:10pm - 5:00pm | Sheena Campbell (library) + in-class work | <b>MIDTERM LAUNCH:</b> literature search tools, then in-class work on the review — <b>bring a laptop</b> (no lecture) | Reflection                                                             |
| 4                    | 8         | Tuesday, October 20th        | 3:10pm - 5:00pm | Dr. Mitchell Feldmann                     | Quantitative genetics: predicting phenotype from genotype                                                             | Reflection & <b>Quiz 3</b> (Reproduction & domestication)              |
| 4                    | 9         | Thursday, October 22nd       | 3:10pm - 5:00pm | Mars (meet at Core Greenhouse Complex)    | Cacao diversity!                                                                                                      | <b>Midterm Topic Explorer</b>                                          |
| 5                    | 10        | Tuesday, October 27th        | 3:10pm - 5:00pm | Grey                                      | Forward genetics: the original gene discovery — mutant screens, the ABC model & MutMap                                | Reflection & <b>Quiz 4</b> (Quantitative genetics)                     |
| 5                    | 11        | Thursday, October 29th       | 3:10pm - 5:00pm | Grey                                      | GWAS/QTL mapping: connecting genetics to phenotypes at large scales                                                   | Group preference survey                                                |
| 6                    | 12        | Tuesday, November 3rd        | 3:10pm - 5:00pm | Grey                                      | Finding the genes: why causal genes are hard to find (heterogeneity, population structure & the Cape Verde story)     | Reflection & <b>Quiz 5</b> (Forward genetics & GWAS)                   |
| 6                    | 13        | Thursday, November 5th       | 3:10pm - 5:00pm | Grey                                      | Climate adaptation: a 400 year history of research                                                                    | Reflection                                                             |
| 7                    | 14        | Tuesday, November 10th       | 3:10pm - 5:00pm | SCOPE graduates                           | SCOPE: student-led plant breeding programs at UC Davis                                                                | Reflection & <b>MIDTERM</b>                                            |
| 7                    | —         | Wednesday, November 11th     | ----            | <b>No class</b>                           | Veterans Day (Wednesday — no Tue/Thu impact)                                                                          | ----                                                                   |
| 7                    | 15        | Thursday, November 12th      | 3:10pm - 5:00pm | Dr. Chaehee Lee                           | The long and short of genomics: what does it mean to “Sequence” something?                                            | Reflection                                                             |
| 8                    | 16        | Tuesday, November 17th       | 3:10pm - 5:00pm | Bayer                                     | Professions in Plant Science and Genetics                                                                             | Reflection & “ <b>Quiz 7</b> ” (group check-in + late-quarter content) |
| 8                    | 17        | Thursday, November 19th      | 3:10pm - 5:00pm | UC Davis Graduate students                | Research projects in Plant Genetics                                                                                   | Reflection                                                             |
| 9                    | 18        | Tuesday, November 24th       | ----            | <b>No class</b>                           | Groups: working on your final presentation                                                                            | Final group check-in “ <b>Quiz 8</b> ”                                 |
| 9                    | 19        | Thursday, November 26th      | ----            | <b>No class</b>                           | Thanksgiving break                                                                                                    | ----                                                                   |
| 10                   | 20        | Tuesday, December 1st        | 3:10pm - 5:00pm | Student presentations                     | FINAL Presentations part 1                                                                                            | Final presentation file & Contributions doc                            |
| 10                   | 21        | Thursday, December 3rd       | 3:10pm - 5:00pm | Student presentations                     | FINAL Presentations part 2                                                                                            | ----                                                                   |
| <b>“finals week”</b> | <b>NA</b> | <b>Friday, December 11th</b> | <b>by 5:00</b>  | ----                                      | ----                                                                                                                  | <b>Extra credit memes due</b>                                          |

Midterm: paper mini literature review/synthesis



# Capstone Final: Group Presentation



# PLS 152: Plant Genetics learning objectives

Fluency in the language of genetics



Connect genetics concepts  
with real-world applications

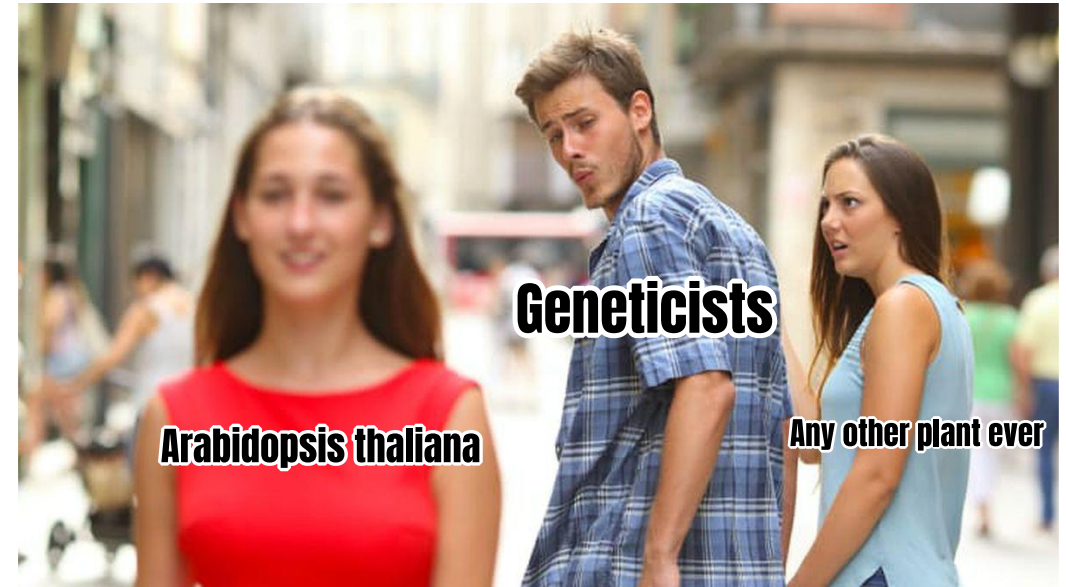
# #1 goal of the course: plant genetics memes



semidwarf-1



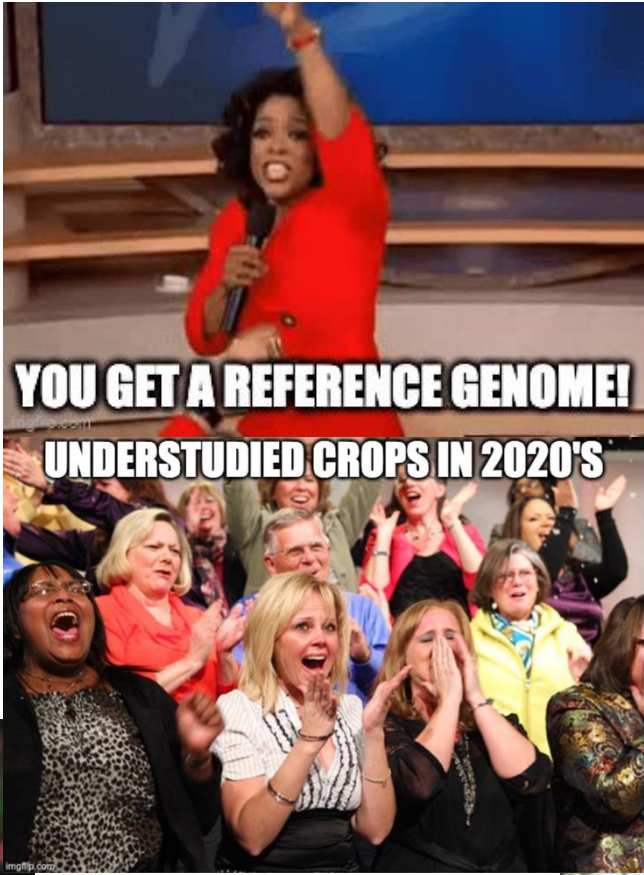
wild type



Realizing there is a difference between  
homologous and homeologous



mRNA:



**Next week:**

**\*Philosophers and scientists struggling  
to understand genetics for millennia\***

**Some Czech monk with a bunch of pea-plants:**



**Transmission genetics: Mendel's rules**